



LIVING STONES FOUNDATION /

APPLIED DATA & DIGITAL INNOVATION LAB

(LATAM)

Technology Standards for Digital Project Development

Purpose

This document establishes the technology standards used by Livingston Foundation for the development of digital platforms, monitoring tools, information management systems, dashboards, artificial intelligence solutions, and projects focused on sustainability, economic development, energy, agriculture, innovation, and social impact.

Its purpose is to ensure technical consistency, maintainability, scalability, and compatibility across projects developed by volunteers, collaborators, and technical teams.

1. General Principles

All projects developed for Livingston Foundation must adhere to the following principles:

- Responsive design for desktop, tablet, and mobile devices.
- Modular and maintainable code structure.
- User-friendly interfaces for both technical and non-technical users.
- Performance optimization.
- Security best practices.
- Functional scalability.

- Reusability of components and development resources.
- Efficient use of development tools and artificial intelligence technologies.

2. Core Technologies

The current technology stack commonly used within Livingston Foundation includes:

Frontend

- HTML5
- CSS3
- JavaScript
- Tailwind CSS
- JSON for data exchange and structured information management

Backend

Depending on project requirements:

- PHP
- JavaScript
- Third-party APIs
- Artificial Intelligence services

Databases

Primary database technology currently used:

- MySQL

Projects should be designed in a way that allows future integration with additional data sources when required.

3. Artificial Intelligence, Multi-Agent Integrations, and Development Tools

Artificial Intelligence represents a core component of Livingston Foundation's innovation ecosystem. Projects may leverage AI technologies to enhance productivity, automate workflows, generate insights, support research, improve decision-making processes, and accelerate software development.

Authorized AI Platforms

Volunteers and developers may utilize AI platforms including, but not limited to:

- OpenAI (ChatGPT)
- Google Gemini
- Anthropic Claude
- DeepSeek
- Perplexity

Developers may also use other AI solutions or technologies of their choice, provided that project security, privacy, intellectual property requirements, and quality standards are maintained.

Multi-Agent Artificial Intelligence Architecture

The innovation core of Livingston Foundation's platforms may rely on a diversified network of Artificial Intelligence models orchestrated through a multi-agent architecture.

Authorized AI Agents and APIs

Project ecosystems may integrate APIs from:

- OpenAI (ChatGPT)
- Google Gemini
- Anthropic Claude
- DeepSeek
- Perplexity

These AI agents can be utilized for:

- Dynamic data analysis
- Knowledge extraction
- Automated diagnostics
- Content generation
- Research support
- Workflow automation
- Decision-support systems
- Sustainability and ESG assessments
- Predictive analysis and recommendations

The selection of models may vary depending on project objectives, performance requirements, cost efficiency, and specific use cases.

Data Synchronization and Persistence

Information generated by AI models shall be processed through the PHP application layer and stored using a combination of:

- MySQL database structures
- JSON objects and data repositories

This hybrid architecture enables:

- Efficient data management
- Scalable knowledge repositories
- Flexible information structures
- Faster API integrations
- Improved system performance
- Long-term persistence of AI-generated insights

The objective is to ensure that project knowledge bases continue expanding while maintaining operational efficiency, system responsiveness, and data consistency.

Responsible Use of AI

Artificial Intelligence tools should be considered productivity enhancers and decision-support systems rather than replacements for human review.

All AI-generated outputs should be subject to:

- Human validation
- Technical review
- Functional testing
- Quality assurance processes
- Security verification when applicable

Final responsibility for project deliverables remains with the development team and project stakeholders.

4. Data Management

All systems should:

- Use structured data formats.
- Implement JSON-based data exchange whenever applicable.
- Support data import and export processes.
- Facilitate integration with external APIs.
- Maintain consistency in data structures and naming conventions.

5. User Experience (UX/UI)

All projects must provide a professional and consistent user experience.

Minimum Requirements

- Fully responsive design.
- Compatibility with modern browsers.
- Mobile-first considerations whenever applicable.
- Intuitive navigation.
- Accessibility best practices.
- Optimized performance across desktops, tablets, and smartphones.

Responsive functionality across all device types is considered a mandatory requirement for every Livingston Foundation project.

6. Version Control

All projects must utilize:

Repository Platform

- GitHub

Best Practices

- Git-based version control.
- Clear documentation of significant updates.
- Source code backup and repository maintenance.
- Organized branch management when appropriate.

7. Infrastructure and Hosting

Projects should be designed for deployment within environments compatible with cPanel hosting.

Requirements

- Compatibility with shared hosting or VPS environments managed through cPanel.
- Simple deployment procedures.
- Linux server compatibility.

- Proper management of files, databases, backups, and application resources.

Development teams should avoid dependencies that require specialized infrastructure unless approved by project leadership.

8. Security

All projects should:

- Protect credentials through secure configuration files.
- Restrict administrative access.
- Utilize HTTPS connections.
- Validate user input and forms.
- Prevent exposure of API keys and sensitive credentials on the frontend.
- Follow basic cybersecurity best practices.

9. Documentation

Each project must include:

Functional Documentation

- Project objectives.
- Target users.
- Functional scope.

Technical Documentation

- Folder structure.
- Installation requirements.
- Database configuration.
- API integrations.
- Deployment procedures.

README File

Every repository must contain a README file with setup instructions, project overview, and maintenance information.

10. Reusability and Standardization

Whenever possible, teams should reuse:

- UI components.
- Visual templates.
- AI integrations.
- Authentication modules.
- Reporting systems.
- Internal libraries and utilities.
- Existing project components and codebases.

This approach promotes faster development cycles, consistency across projects, and more efficient knowledge transfer among volunteers and contributors.

Examples:

ENERGYTRANSIT

<https://arletdesign.digital/energytransit/>

GREEN VERDE

<https://livingstonesglobal.online/greenverdehub/>

<https://livingstonesglobal.online/greenverdeco2/arkansas.php>

EQUINOX CORE

<https://livingstonesglobal.online/equinoxcore/>

CULTIVECONNECT - CAPSTONE PROJECT

<https://livingstonesglobal.online/cultiveconnect/>

SONAR CORE AI

<https://livingstonesglobal.online/sonarcore/>



Newly Launched Projects

Lunmenindex

<https://livingstonesglobal.online/lumenindex/>

Final Consideration

These standards reflect the technologies, methodologies, and development practices currently utilized by Livingston Foundation.

Any significant deviation from these standards or adoption of new technologies should be documented and justified to ensure compatibility, maintainability, security, and long-term sustainability across the Foundation's digital ecosystem.

Research Data Sources and Open Data Resources

Many Livingston Foundation initiatives focus on Latin America and the Caribbean. Volunteers and developers, particularly those who work primarily in English, are encouraged to leverage international open-data repositories and public statistical resources that provide high-quality datasets for research, analysis, visualization, artificial intelligence applications, monitoring systems, and decision-support platforms.

The following sources are recommended as primary references:

World Bank Open Data

The World Bank provides one of the most comprehensive collections of global development indicators, including extensive datasets covering Latin America and the Caribbean.

Areas of coverage include:

- Economic development
- Poverty and inequality

- Education
- Energy
- Agriculture
- Climate change
- Infrastructure
- Governance
- Sustainable Development Goals (SDGs)

Website:

<https://data.worldbank.org/>

World Health Organization (WHO)

The World Health Organization provides access to global health indicators through the Global Health Observatory (GHO).

Areas of coverage include:

- Public health
- Disease surveillance
- Healthcare systems
- Mortality and life expectancy
- Health-related Sustainable Development Goals

Website:

<https://www.who.int/data/gho/info/gho-odata-api>

Institute for Health Metrics and Evaluation (IHME)

IHME provides extensive datasets and analytical resources related to global health, disease burden, population health trends, and healthcare outcomes.

Areas of coverage include:

- Global Burden of Disease (GBD)
- Health risk factors
- Mortality and morbidity data
- Health forecasting and modeling

Website:

<http://ghdx.healthdata.org/>

Pan American Health Organization (PAHO)

PAHO provides regional health data for countries and territories across the Americas through its open-data initiatives.

Areas of coverage include:

- Public health
- Epidemiology
- Healthcare access
- Regional development indicators
- Health policy analysis

Website:

<https://www.paho.org/data/>

Statistical Information System Collaboration Community (SIS-CC)

SIS-CC is an international open-source community supporting statistical organizations through modern data standards and official statistics frameworks.

Areas of coverage include:

- Official statistics
- SDMX standards
- Statistical modernization
- Data lifecycle management
- Open government data initiatives

Website:

<https://siscc.org/>

Open Data and AI Utilization

Volunteers are encouraged to combine these public datasets with Artificial Intelligence tools such as ChatGPT, Gemini, Claude, DeepSeek, and Perplexity to:

- Conduct research and policy analysis.
- Generate insights and summaries.
- Create dashboards and visualizations.
- Develop predictive models.
- Support ESG and sustainability initiatives.
- Build knowledge repositories and decision-support systems.

All data sources should be properly cited, validated, and reviewed prior to inclusion in production systems or public-facing reports.

Data Extraction, Cleaning, and AI-Assisted Development Tools

Livingston Foundation encourages volunteers, researchers, analysts, and developers to leverage Artificial Intelligence and modern data engineering tools to accelerate the extraction, cleaning, transformation, analysis, and visualization of structured and unstructured data.

These tools support projects related to sustainability, ESG, public policy, climate resilience, agriculture, health, economic development, and social innovation.

AI Platforms for Research and Data Processing

Volunteers may utilize Artificial Intelligence platforms including:

Platform	Website
OpenAI (ChatGPT)	https://chatgpt.com/
Google Gemini	https://gemini.google.com/
Anthropic Claude	https://claude.ai/
DeepSeek	https://www.deepseek.com/

Perplexity <https://www.perplexity.ai/>

Kimi AI <https://kimi.ai/>

Bolt.new <https://bolt.new/>

These tools may be used for:

- Data extraction and summarization
- Data cleaning recommendations
- Metadata generation
- Data transformation workflows
- Research and literature reviews
- Statistical interpretation
- Dashboard and report generation
- Software development assistance
- Documentation and technical writing

AI-generated outputs should always be reviewed and validated by the project team before being incorporated into production systems or published deliverables.

OpenRefine

OpenRefine is the preferred open-source tool for cleaning and transforming large datasets.

Website

<https://openrefine.org/>

Tutorial

<https://youtu.be/D-HAMsER4yE?feature=shared>

Recommended Uses

- Data standardization
- Duplicate detection
- Clustering and reconciliation
- CSV, Excel, and JSON processing
- Public data preparation
- Data imported from APIs and external sources

Python + Pandas using Google Colab

Google Colab provides free cloud-based notebook environments accessible through any browser.

Website

<https://colab.research.google.com/>

Recommended Libraries

- Pandas
- NumPy
- Scikit-Learn
- Sweetviz
- AutoViz
- sklearn.impute

Recommended Uses

- Data cleaning
- Missing value treatment
- Statistical analysis
- Machine learning preparation
- Dataset transformation
- Automated reporting

Visual Studio Code (VS Code)

Visual Studio Code is the recommended development environment for software and data projects.

Website

<https://code.visualstudio.com/>

Features

- Python support
- GitHub integration
- Jupyter Notebook compatibility
- Debugging tools
- AI-assisted coding extensions
- Multi-language support

Replit

Cloud-based collaborative development environment.

Website

<https://replit.com/>

Recommended Uses

- Python development
- Rapid prototyping
- Collaborative coding
- Data analysis projects
- Educational and volunteer initiatives

Bolt.new

AI-powered full-stack application builder.

Website

<https://bolt.new/>

Tutorial

<https://youtu.be/f-rMQzpPWlY?feature=shared>

Features

- AI-assisted application generation
- Full-stack development workflows
- Browser-based coding environment
- Rapid prototyping
- Exportable source code

V0.dev

AI-powered platform for generating web interfaces and application prototypes.

Website

<https://v0.dev/>

Recommended Uses

- UI prototyping
- Dashboard design

- Frontend concepts
- Rapid design validation

Additional Data Engineering and ETL Tools

DataCleaner

Data quality and cleansing platform.

Website:

<https://datacleaner.org/>

Talend Open Studio

ETL, data integration, and transformation platform.

Website:

<https://www.talend.com/>

KNIME Analytics Platform

Open-source analytics and workflow automation platform.

Website:

<https://www.knime.com/>

Trifacta Wrangler

Data preparation and transformation platform.

Website:

<https://www.trifacta.com/>

Google Cloud Dataprep

Cloud-based data preparation platform powered by Trifacta.

Website:

<https://cloud.google.com/dataprep>

Pandas

Python library for data manipulation and analysis.

Website:

<https://pandas.pydata.org/>

AI-Assisted Programming Platforms

Depending on project requirements, volunteers may also utilize:

Platform	Website
Windsurf	https://windsurf.com/
Cursor	https://cursor.com/
Blackbox AI	https://www.blackbox.ai/
Replit Ghostwriter	https://replit.com/
V0.dev	https://v0.dev/
Bolt.new	https://bolt.new/
OpenAI (ChatGPT)	https://chatgpt.com/
Google Gemini	https://gemini.google.com/
Anthropic Claude	https://claude.ai/

These platforms can assist with:

- Software development
- Data engineering and Data cleaning
- Dashboard creation
- API integrations
- Documentation generation
- Rapid prototyping
- AI-assisted coding

Tool Selection Philosophy

Livingston Foundation promotes flexibility, innovation, and responsible use of Artificial Intelligence.

Volunteers are encouraged to select the tools that best fit their expertise and project objectives while maintaining:

- Data quality
- Reproducibility
- Documentation standards
- Security requirements
- Long-term maintainability

The goal is not to enforce a single toolset, but rather to provide a recommended ecosystem that accelerates innovation, research, software development, and evidence-based decision-making throughout Livingston Foundation initiatives.